

MVP Feature List

1. Image Quality Check

- Basic blur/lighting/resolution check; flag poor-quality uploads

2. DR Diagnostic Model (degradation-trained)

- Fine-tuned EfficientNet-B0 on APTOS dataset
- Fixed degradation augmentation (downscale, blur, compression, brightness — one config, no tuning)
- Output: DR present/absent + severity grade (0–4)

3. Grad-CAM Heatmap

- Visual overlay showing regions driving the prediction

4. Confidence Flagging (simplified)

- Softmax-threshold based confidence score
- Auto-flag low-confidence predictions for review

5. Doctor Portal

- Login, upload image, view prediction + heatmap + confidence
- Approve/reject (no granular editing)
- Manage/view patient list

6. Patient Portal

- Login, view own report (image, heatmap, severity, doctor notes)
- **LLM-based explanation (lite version):** single prompt combining structured report data (severity, confidence) + a short doctor-reviewed reference blurb per grade → generates a plain-language explanation
- **Multilingual output:** same LLM call generates explanation in English or Hindi (one multilingual-capable model, no separate translation pipeline)
- *(No live notifications — report just appears when ready)*

7. Report Generation

- Templated HTML/PDF: image + heatmap + severity + confidence + doctor sign-off

Explicitly cut from this MVP (v2 backlog)

- MC Dropout / full uncertainty calibration
- Offline/on-device inference
- **Full RAG pipeline** (vector DB, embedding index, retrieval over a curated medical knowledge base — MVP uses a lightweight single-prompt version instead)
- **Voice AI Assistant (STT/TTS)** — MVP is text-only explanation, no speech input/output
- **Multilingual support beyond Hindi/English** — MVP covers just these two languages
- AI-generated (non-templated) *doctor-facing* reports — the doctor's report itself stays templated; LLM explanation is patient-facing only
- Doctor edit workflow on AI outputs
- Patient notifications
- Deep ensembles